

COMPLEXITY-SENSITIVE ADDITIVE ENERGY AND OFF-DIAGONAL YOUNG INEQUALITIES ON BOUNDED-DEGREE ALGEBRAIC VARIETIES

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ABSTRACT. We develop additive-energy estimates and weighted Young inequalities for finite sets on bounded-degree real algebraic varieties. For an irreducible m -dimensional variety V , let

$$\sigma(V) = 2m - \dim \overline{V - V}^{\text{Zar}}, \quad \alpha(V) = \max \left\{ 2, 1 + \frac{2\sigma(V)}{m} \right\}.$$

For every $a \in [\alpha(V), 3)$ we define a finite-degree translation-partition flag parameter $\Lambda_{a,R}(X; V)$ and prove

$$E(X) \ll \Lambda_{a,R}(X; V)^{3-a} |X|^{a+\varepsilon}.$$

This recovers the line-concentration theorem of Jing and Wu for algebraic surfaces in $\mathbb{R}^3$. For codimension-two quadratic threefolds

$$\{(u, Q_1(u), Q_2(u)) : u \in \mathbb{R}^3\} \subset \mathbb{R}^5$$

with positive-definite Q_1 and simple generalized spectrum, we prove the sharp estimate $E(X) \ll_\varepsilon |X|^{2+\varepsilon}$ without a flag loss. Hereditary versions of these estimates imply weighted L^4 restriction bounds and off-diagonal Young inequalities; at the near-diagonal threshold the sharp region is

$$1 \leq p, q \leq 2, \quad p^{-1} + q^{-1} \geq 1.$$

We also prove a sharp turning-complexity extension of the Cushman–Demeter–Wu theorem: $J_3(P) \ll_\varepsilon \kappa(P)^2 |P|^{3+\varepsilon}$, with matching examples at every power scale.

1. INTRODUCTION

1.1. Additive energy and geometric support restrictions. Let G be a torsion-free abelian group and let $X \subset G$ be finite. The additive energy of X is

$$(1.1) \quad E(X) = \sum_{t \in G} r_{X-X}(t)^2, \quad r_{X-X}(t) = \#\{(x, y) \in X^2 : x - y = t\}.$$

It satisfies the universal bounds

$$2|X|^2 - |X| \leq E(X) \leq |X|^3.$$

Date: Working draft, August 19, 2026.

2020 Mathematics Subject Classification. Primary 11B30, 42B10; Secondary 14P10, 52C10, 26D15.

Key words and phrases. additive energy, algebraic varieties, polynomial partitioning, Young's inequality, difference varieties, convex curves.

The lower bound is the diagonal scale; the upper bound is attained, up to constants, by arithmetic progressions. If G_0 denotes the subgroup generated by X , then $G_0 \cong \mathbb{Z}^r$ for some r , and Plancherel gives

$$(1.2) \quad E(X) = \int_{\mathbb{T}^r} \left| \widehat{\mathbf{1}_X}(\xi) \right|^4 d\xi.$$

Thus a near-diagonal energy theorem is simultaneously a discrete L^4 restriction theorem.

Curvature and algebraic complexity can suppress additive multiplicity. Bourgain and Demeter related such questions to decoupling and discrete restriction [6]. Jing and Wu recently proved that if $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ is irreducible of degree at least two, then every finite $X \subset Z(F)$ satisfies

$$(1.3) \quad E(X) \ll_{\deg V, \varepsilon} \Lambda_V(X) |X|^{2+\varepsilon}, \quad \Lambda_F(X) = \max \left\{ 1, \sup_{\ell \subset Z(F)} |X \cap \ell| \right\},$$

where the supremum is over affine lines contained in the surface [11]. Their theorem resolves the near-diagonal problem for arbitrary algebraic surfaces and identifies concentration on surface-contained lines as the only obstruction.

There is a related higher-order phenomenon on curves. For a finite $P \subset \mathbb{R}^2$, define

$$(1.4) \quad J_3(P) = \#\{(p_1, \dots, p_6) \in P^6 : p_1 + p_2 + p_3 = p_4 + p_5 + p_6\}.$$

Cushman, Demeter, and Wu proved that $J_3(P) \ll_\varepsilon |P|^{3+\varepsilon}$ whenever P lies on a strictly convex graph [8]. Their proof combines an interlacing structure inside equal-sum fibers with Fourier orthogonality for ordered chord cones.

The present paper has two goals. The first is to identify the dimension and translation-complexity parameters that govern the polynomial-partitioning argument on general algebraic varieties. The second is to convert the resulting set estimates into weighted and off-diagonal Young inequalities, in a form that records the sharp critical exponents. This functional viewpoint is in the tradition of the sharp Young and Fourier inequalities of Beckner and Brascamp–Lieb [4, 7], while the geometric decomposition follows the polynomial-partitioning philosophy initiated by Guth and Katz [10] and developed over general varieties by Walsh.

1.2. Difference varieties and the intrinsic exponent. Let $V \subset \mathbb{R}^n$ be an irreducible real algebraic variety whose complexification $V_{\mathbb{C}}$ is irreducible and whose real locus is Zariski dense. Put

$$m = \dim V_{\mathbb{C}}, \quad \mathcal{D}(V) = \overline{V_{\mathbb{C}} - V_{\mathbb{C}}}^{\text{Zar}} \subset \mathbb{C}^n, \quad d(V) = \dim \mathcal{D}(V).$$

The difference map

$$\delta_V : V_{\mathbb{C}} \times V_{\mathbb{C}} \longrightarrow \mathcal{D}(V), \quad \delta_V(x, y) = x - y,$$

has generic fiber dimension

$$(1.5) \quad \sigma(V) = 2m - d(V).$$

Accordingly, for a generic difference t , the translate intersection $V_{\mathbb{C}} \cap (V_{\mathbb{C}} + t)$ has dimension $\sigma(V)$. The exponent produced by two applications of the crossing estimate is

$$(1.6) \quad \alpha(V) = \max \left\{ 2, 1 + \frac{2\sigma(V)}{m} \right\}.$$

The diagonal contribution forces the first term. The second term is the fixed point of the polynomial-partition recurrence.

If $V^m \subset \mathbb{R}^{m+c}$ has maximal difference dimension, then

$$d(V) = \min(2m, m+c), \quad \sigma(V) = \max(0, m-c),$$

and hence

$$(1.7) \quad \alpha_{m,c} = \max \left\{ 2, 3 - \frac{2c}{m} \right\}.$$

The line $m = 2c$ is the critical boundary of the near-diagonal range. The surface case $V^2 \subset \mathbb{R}^3$ and the codimension-two cases $V^3 \subset \mathbb{R}^5$ and $V^4 \subset \mathbb{R}^6$ lie on or below this threshold.

1.3. The flagged energy theorem. A general variety can contain affine rulings, positive-dimensional translation stabilizers, and lower-dimensional subvarieties with a worse intrinsic exponent. These phenomena prevent a uniform parameter-free theorem. We therefore introduce, in [section 3.1](#), a finite-degree translation-partition flag parameter $\Lambda_{a,R}(X; V)$. It records:

- (i) concentration on cosets of translation stabilizers;
- (ii) concentration on translate intersections whose dimension exceeds the generic value;
- (iii) concentration on bounded-degree flag subvarieties whose intrinsic exponent is larger than the target exponent.

The parameter is monotone under passage to subsets and satisfies $1 \leq \Lambda_{a,R}(X; V) \leq |X|$.

Our main general theorem is the following. The algebraic conventions and the precise definition of $\Lambda_{a,R}$ are given in [sections 2.2](#) and [3.1](#).

Theorem 1.1. *Let $n, \Delta \geq 1$, let $2 \leq a < 3$, and let $\varepsilon > 0$. There are constants*

$$R = R(n, \Delta, a, \varepsilon), \quad C = C(n, \Delta, a, \varepsilon)$$

with the following property. Let $V \subset \mathbb{R}^n$ be an admissible irreducible algebraic variety of degree at most Δ such that $\alpha(V) \leq a$. Then every finite $X \subset V$ satisfies

$$(1.8) \quad E(X) \leq C \Lambda_{a,R}(X; V)^{3-a} |X|^{a+\varepsilon}.$$

Taking $a = \alpha(V)$ gives the intrinsic form whenever $\alpha(V) < 3$. The endpoint $\alpha(V) = 3$ is governed by the universal trivial estimate $E(X) \leq |X|^3$ and has no contractive partition recurrence. When V is an irreducible nonplanar surface in $\mathbb{R}^3$, the flag parameter is comparable, with degree-dependent constants, to $\Lambda_F(X)$ in [\(1.3\)](#); see [theorem 3.4](#). Thus [theorem 1.1](#) recovers

the theorem of Jing and Wu. Its proof follows their two-crossing mechanism, but runs on a spatial flag of partition walls and uses the difference variety to determine the correct fiber dimension.

The power $3 - a$ has the expected interpolation form. If a base example has energy of order N^a and one takes its Cartesian product with an arithmetic progression of length L , then the resulting energy is of order $L^{3-a}N^a$. In particular, the power is sharp in the classes for which the base exponent is attained.

1.4. A sharp codimension-two threefold theorem. The flag parameter in [theorem 1.1](#) is deliberately safe. It can overcount exceptional translate fibers that have large individual occupancy but small total energy. The first nontrivial example where this distinction matters is a three-dimensional quadratic graph in $\mathbb{R}^5$.

Let $Q_j(u) = u^\top A_j u$ be quadratic forms on $\mathbb{R}^3$. We say that (Q_1, Q_2) has simple generalized spectrum if A_1 is positive definite and $A_1^{-1/2} A_2 A_1^{-1/2}$ has three distinct eigenvalues.

Theorem 1.2. *Let*

$$\Sigma = \{(u, Q_1(u), Q_2(u)) : u \in \mathbb{R}^3\} \subset \mathbb{R}^5,$$

where (Q_1, Q_2) has simple generalized spectrum. Then, for every $\varepsilon > 0$ and every finite $X \subset \Sigma$,

$$(1.9) \quad E(X) \ll_\varepsilon |X|^{2+\varepsilon}.$$

The exponent 2 is sharp.

After linear changes of variables, the model becomes

$$\Phi(u) = (u, |u|^2, \lambda_1 u_1^2 + \lambda_2 u_2^2 + \lambda_3 u_3^2), \quad \lambda_i \neq \lambda_j.$$

For a generic difference, the parameter fiber is an affine line. It becomes a plane only when the parameter displacement lies in one of the three eigendirections. A Gram-matrix argument shows that the total energy of all these rank-drop directions is $O(N^2)$. Polynomial partitioning in the parameter space produces $O(D^3)$ cells, while the two crossing estimates cost only D^2 ; at exponent 2 the cellular recursion therefore gains a genuine factor D^{-1} . The two-dimensional wall is reduced to the Jing–Wu theorem by a finite Freiman projection to $\mathbb{R}^3$.

The simple-spectrum hypothesis cannot be removed without replacing it by a geometric nondegeneracy condition. If Q_2 is proportional to Q_1 , the graph lies in an affine copy of a three-dimensional paraboloid in $\mathbb{R}^4$, where the universal threshold is $N^{7/3}$ rather than N^2 .

1.5. Weighted and off-diagonal Young inequalities. A hereditary set-energy theorem has a functional consequence. If every $A \subset X$ satisfies

$$E(A) \ll K^{3-a} |A|^{a+o(1)},$$

then

$$(1.10) \quad \|\widehat{f}\|_{L^4} \ll K^{(3-a)/4} |X|^{o(1)} \|f\|_{\ell^{4/a}}, \quad \text{supp } f \subset X.$$

Combining two such estimates with Plancherel gives a geometric Young endpoint. Interpolation with the classical endpoints $\ell^1 * \ell^2 \rightarrow \ell^2$ and $\ell^2 * \ell^1 \rightarrow \ell^2$ yields a full off-diagonal family.

The near-diagonal case $a = 2$ is especially clean.

Theorem 1.3. *Suppose finite sets X, Y satisfy hereditary estimates*

$$E(A) \ll_{\eta} K_X |A|^{2+\eta} \quad (A \subset X), \quad E(B) \ll_{\eta} K_Y |B|^{2+\eta} \quad (B \subset Y)$$

for every $\eta > 0$. Let $1 \leq p, q \leq 2$ and assume

$$\frac{1}{p} + \frac{1}{q} \geq 1.$$

Put

$$\theta = \max \left\{ 0, 3 - 2 \left(\frac{1}{p} + \frac{1}{q} \right) \right\}.$$

Then for every $\varepsilon > 0$ and all f, g supported on X, Y ,

$$(1.11) \quad \|f * g\|_{\ell^2} \ll_{p,q,\varepsilon} (K_X K_Y)^{\theta/4} (|X||Y|)^{\varepsilon} \|f\|_{\ell^p} \|g\|_{\ell^q}.$$

The exponent region is sharp up to the subpolynomial loss. If the class contains arithmetic progressions on affine rulings, the joint power of $K_X K_Y$ is also sharp.

For the threefold in [theorem 1.2](#), one has $K_X = K_Y = 1$. For algebraic surfaces in $\mathbb{R}^3$, one may take $K_X = \Lambda_F(X)$ and $K_Y = \Lambda_G(Y)$. Thus the same off-diagonal theorem simultaneously applies to the recent results of Jing–Wu and to the new codimension-two case.

1.6. A sharp turning-complexity theorem for three-fold curve energy. Our final main result is a parallel complexity theorem for [\(1.4\)](#). Fix a linear functional ℓ that is injective on $P \subset \mathbb{R}^2$ and order

$$\ell(p_1) < \dots < \ell(p_N).$$

Choose a complementary linear functional m and form the adjacent slopes

$$s_i = \frac{m(p_{i+1}) - m(p_i)}{\ell(p_{i+1}) - \ell(p_i)}.$$

The least number of consecutive point blocks on each of which the corresponding slope sequence is strictly increasing or strictly decreasing is denoted by $\kappa_{\ell}(P)$. This number does not depend on the choice of m .

Theorem 1.4. *For every $\varepsilon > 0$,*

$$(1.12) \quad J_3(P) \ll_{\varepsilon} \kappa_{\ell}(P)^2 |P|^{3+\varepsilon}.$$

Moreover, for every pair of integers $K, n \geq 1$, there is a set P with $|P| = Kn$ for which

$$\kappa_{\ell}(P) \asymp K, \quad J_3(P) \asymp K^2 |P|^3.$$

Thus the power κ^2 is sharp at every intermediate power scale.

The proof partitions into signed-convex leaves and uses one global L^6 strip projection estimate. The strips charge the leaves only once; a repeated node-by-node triangle inequality would lose an additional factor of κ . Each leaf is interpolated by a strictly convex or strictly concave graph and is handled by the theorem of Cushman–Demeter–Wu.

1.7. Organization. Basic Fourier and algebraic tools are collected in [section 2](#). The flagged energy theorem is proved in [section 3](#). The codimension-two quadratic threefold is treated in [section 4](#). Weighted and off-diagonal inequalities occupy [section 5](#). The turning-complexity theorem is proved in [section 6](#). Sharpness, higher-dimensional consequences, and the remaining geometric difficulties are discussed in [section 7](#).

2. PRELIMINARIES

2.1. Fourier identities on finite supports. Every finite subset of $\mathbb{R}^n$ is contained in a finitely generated torsion-free subgroup, hence in a group isomorphic to $\mathbb{Z}^r$. We freely identify such a subgroup with $\mathbb{Z}^r$ when using Fourier analysis. This does not change any additive relation.

For a finitely supported function f on an abelian group, put

$$\tilde{f}(x) = \overline{f(-x)}, \quad R_f(t) = (f * \tilde{f})(t).$$

For finite sets A, B , write

$$r_{A-B}(t) = \#\{(a, b) \in A \times B : a - b = t\},$$

and define the mixed energy

$$(2.1) \quad E(A, B) = \sum_t r_{A-B}(t)^2.$$

Thus $E(A) = E(A, A)$.

Lemma 2.1. *Let f, g be finitely supported. Then*

$$(2.2) \quad \|f * g\|_{\ell^2}^2 = \sum_t R_f(t) \overline{R_g(t)} = \int |\hat{f}|^2 |\hat{g}|^2,$$

$$(2.3) \quad \|\hat{f}\|_{L^4}^4 = \sum_t |R_f(t)|^2.$$

In particular,

$$(2.4) \quad E(A, B) = \sum_t r_{A-A}(t) r_{B-B}(t).$$

If $A = \bigsqcup_{j=1}^J A_j$, then

$$(2.5) \quad E(A)^{1/4} \leq \sum_{j=1}^J E(A_j)^{1/4}.$$

Proof. Identify the generated subgroup with $\mathbb{Z}^r$ and use Fourier series on $\mathbb{T}^r$. Equations (2.2) and (2.3) follow from Plancherel. Equation (2.4) is the indicator-function case of (2.2). Finally,

$$E(A)^{1/4} = \|\widehat{\mathbf{1}_A}\|_{L^4} \leq \sum_j \|\widehat{\mathbf{1}_{A_j}}\|_{L^4} = \sum_j E(A_j)^{1/4}.$$

□

We shall also use the elementary bounds

$$(2.6) \quad 2|A|^2 - |A| \leq E(A) \leq |A|^3.$$

2.2. Algebraic conventions. An *admissible irreducible real algebraic variety* $W \subset \mathbb{R}^n$ means the real locus of an irreducible complex affine variety $W_{\mathbb{C}} \subset \mathbb{C}^n$ defined over $\mathbb{R}$, with W Zariski dense in $W_{\mathbb{C}}$. We set

$$\dim W = \dim_{\mathbb{C}} W_{\mathbb{C}}, \quad \deg W = \deg W_{\mathbb{C}}.$$

General bounded-degree real algebraic sets admit a bounded real-algebraic stratification: after separating conjugate complex components and their lower-dimensional real intersections, the real locus is covered by $O_{n,\Delta}(1)$ admissible irreducible pieces of bounded degree, together with lower-dimensional pieces; see, for example, [3]. Since (2.5) handles finite unions, it is enough to formulate the main argument for admissible irreducible varieties.

For such a W of dimension $k \geq 1$, define

$$(2.7) \quad \mathcal{D}(W) = \overline{W_{\mathbb{C}} - W_{\mathbb{C}}}^{\text{Zar}}, \quad d(W) = \dim \mathcal{D}(W), \quad \sigma(W) = 2k - d(W).$$

The fiber-dimension theorem implies that $\sigma(W)$ is the generic complex dimension of

$$W_{\mathbb{C}} \cap (W_{\mathbb{C}} + t), \quad t \in \mathcal{D}(W).$$

We put

$$(2.8) \quad \alpha(W) = \max \left\{ 2, 1 + \frac{2\sigma(W)}{k} \right\}.$$

The real translation stabilizer is

$$(2.9) \quad H_W = \text{Stab}(W) = \{t \in \mathbb{R}^n : W + t = W\}.$$

Lemma 2.2. *The set H_W is a real linear subspace of $\mathbb{R}^n$.*

Proof. It is plainly an additive subgroup. If $t \in H_W$, then $W + jt = W$ for every $j \in \mathbb{Z}$. For every polynomial P vanishing on $W_{\mathbb{C}}$ and every $x \in W_{\mathbb{C}}$, the polynomial $s \mapsto P(x + st)$ vanishes at all integers, hence identically. Thus $W_{\mathbb{C}} + st = W_{\mathbb{C}}$ for every $s \in \mathbb{C}$, and in particular $st \in H_W$ for every real s . Therefore H_W is a real vector subspace. □

Define the exceptional translation set

$$(2.10) \quad \mathcal{B}(W) = \{t \in \mathbb{R}^n \setminus H_W : \dim_{\mathbb{C}}(W_{\mathbb{C}} \cap (W_{\mathbb{C}} + t)) > \sigma(W)\}.$$

For a finite $A \subset W$, set

$$(2.11) \quad \lambda_W(A) = \max \left\{ 1, \sup_{z \in \mathbb{R}^n} |A \cap (z + H_W)|, \sup_{t \in \mathcal{B}(W)} |A \cap W \cap (W + t)| \right\}.$$

Empty suprema are interpreted as zero.

2.3. Polynomial partitioning and real component bounds. We use two standard consequences of the polynomial method over varieties. The first combines polynomial partitioning on an arbitrary variety with the usual component bound; see Walsh [13] and, in codimension at most two, Basu–Sombra [2].

Theorem 2.3 (Polynomial partitioning on a variety). *Let $W \subset \mathbb{R}^n$ be admissible irreducible of dimension k and degree at most Δ , and let $A \subset W$ be finite. For every integer $D \geq 2$, there is a polynomial P of degree at most D , not vanishing identically on $W_{\mathbb{C}}$, such that every semialgebraically connected component of*

$$W \setminus Z(P)$$

contains at most

$$C_{n,\Delta} \frac{|A|}{D^k}$$

points of A . The number of components meeting A is at most $C_{n,\Delta} D^k$.

The second is a form of the Barone–Basu sign-condition estimate [1].

Theorem 2.4 (Component bound). *Let $Z \subset \mathbb{R}^n$ be a real algebraic set of dimension at most s and degree at most B . Let P_1, P_2 be polynomials of degree at most D . Then*

$$(2.12) \quad b_0\left(Z \setminus (Z(P_1) \cup Z(P_2))\right) \leq C_{n,B}(D+1)^s.$$

We also use standard affine Bezout bounds: intersections of bounded-degree varieties have bounded total degree, and if a line meets an algebraic variety of degree B in more than B points, then the line is contained in the variety. These facts may be found, for example, in [3].

3. THE FLAGGED ENERGY THEOREM

3.1. The flag parameter. Fix an admissible irreducible $V \subset \mathbb{R}^n$, a target exponent $2 \leq a < 3$, and a degree cutoff $R \geq \deg V$. Let $\mathcal{F}_R(V)$ denote the family of positive-dimensional admissible irreducible subvarieties $W \subset V$ with $\deg W \leq R$. Define

$$(3.1) \quad \Lambda_{a,R}(X; V) = \max \left\{ \begin{array}{l} 1, \\ \sup_{\substack{W \in \mathcal{F}_R(V) \\ \alpha(W) \leq a}} \lambda_W(X \cap W), \\ \sup_{\substack{W \in \mathcal{F}_R(V) \\ \alpha(W) > a}} |X \cap W| \end{array} \right\}.$$

The parameter is monotone: if $X' \subset X$, then

$$(3.2) \quad \Lambda_{a,R}(X'; V) \leq \Lambda_{a,R}(X; V).$$

The degree cutoff is essential. Without it, one could fit a high-degree subvariety to a prescribed finite configuration. In the proof of [theorem 1.1](#), R is chosen large enough to contain every descendant produced by the finite partition-wall induction.

3.2. Exceptional translations. For $A \subset W$, write

$$r_A(t) = r_{A-A}(t).$$

Lemma 3.1. *Let W be admissible irreducible and let $A \subset W$ be finite. Then*

$$(3.3) \quad \sum_{t \in H_W \cup \mathcal{B}(W)} r_A(t)^2 \leq 2\lambda_W(A)|A|^2.$$

Proof. Partition A into its intersections A_L with the affine cosets of H_W . If $t \in H_W$, then both points in a representation $x - y = t$ belong to the same coset. Hence

$$\sum_{t \in H_W} r_A(t) = \sum_L |A_L|^2 \leq \lambda_W(A)|A|.$$

Since $r_A(t) \leq |A|$,

$$(3.4) \quad \sum_{t \in H_W} r_A(t)^2 \leq \lambda_W(A)|A|^2.$$

If $t \in \mathcal{B}(W)$, then every first point in a representation of t lies in $A \cap W \cap (W + t)$, so

$$r_A(t) \leq \lambda_W(A).$$

Therefore

$$(3.5) \quad \sum_{t \in \mathcal{B}(W)} r_A(t)^2 \leq \lambda_W(A) \sum_t r_A(t) = \lambda_W(A)|A|^2.$$

Combining (3.4) and (3.5) proves the result. $\square$

The elementary inequality

$$(3.6) \quad \lambda M^2 \leq \lambda^{3-a} M^a, \quad 1 \leq \lambda \leq M, \quad 2 \leq a < 3,$$

will repeatedly convert [theorem 3.1](#) into the target scale.

3.3. The two-crossing estimate. Let $W \subset \mathbb{R}^n$ have dimension k and put $\sigma = \sigma(W)$. Apply [theorem 2.3](#) to $A \subset W$ with a polynomial P of degree D . Let

$$A_0 = A \cap Z(P), \quad Y = A \setminus A_0 = \bigsqcup_i A_i,$$

where the A_i lie in distinct components of $W \setminus Z(P)$.

Proposition 3.2. *With the notation above,*

$$(3.7) \quad E(Y) \leq C_{n, \deg W} \left(D^\sigma \lambda_W(A) |A|^2 + D^{2\sigma} \sum_i E(A_i) \right).$$

Proof. Let

$$\text{Bad}(W) = H_W \cup \mathcal{B}(W).$$

Fix $t \notin \text{Bad}(W)$. The algebraic set

$$W_t = W \cap (W + t)$$

has complex dimension at most σ and degree bounded in terms of $\deg W$. Apply [theorem 2.4](#) to W_t and the two polynomials

$$P(x), \quad P(x - t).$$

It follows that

$$(3.8) \quad \#\{(i, j) : r_{A_i - A_j}(t) > 0\} \leq CD^\sigma.$$

Indeed, on each connected component of

$$W_t \setminus (Z(P) \cup Z(P(\cdot - t))),$$

the points x and $x - t$ remain in fixed partition cells.

By Cauchy–Schwarz and (3.8),

$$(3.9) \quad r_Y(t)^2 \leq CD^\sigma \sum_{i,j} r_{A_i - A_j}(t)^2, \quad t \notin \text{Bad}(W).$$

Summing and using [theorem 3.1](#),

$$(3.10) \quad E(Y) \leq 2\lambda_W(A) |A|^2 + CD^\sigma \sum_{i,j} \sum_t r_{A_i - A_j}(t)^2.$$

By (2.4),

$$(3.11) \quad \sum_{i,j} \sum_t r_{A_i - A_j}(t)^2 = \sum_u \left(\sum_i r_{A_i - A_i}(u) \right)^2.$$

For $u \notin \text{Bad}(W)$, the argument giving (3.8), now with $i = j$, yields

$$\#\{i : r_{A_i - A_i}(u) > 0\} \leq CD^\sigma.$$

Hence

$$(3.12) \quad \left(\sum_i r_{A_i - A_i}(u) \right)^2 \leq CD^\sigma \sum_i r_{A_i - A_i}(u)^2.$$

For $u \in \text{Bad}(W)$, the left side is bounded by $r_A(u)^2$, and [theorem 3.1](#) applies. Thus

$$(3.13) \quad \sum_{i,j} \sum_t r_{A_i - A_j}(t)^2 \leq 2\lambda_W(A) |A|^2 + CD^\sigma \sum_i E(A_i).$$

Substitute (3.13) into (3.10) and absorb lower powers of D . □

3.4. Proof of the flagged theorem.

Proof of [theorem 1.1](#). We prove a slightly stronger target-exponent statement by induction on the dimension. Fix $a \in [2, 3)$ and assume $\alpha(V) \leq a$.

We first choose a finite degree tower. Starting with the top degree bound $\Delta_m = \Delta$, descend in dimension. At dimension k , choose a partition degree D_k sufficiently large in terms of $n, \Delta_k, a, \varepsilon$ that the cellular factor in [\(3.16\)](#) below is at most $1/100$. Then choose Δ_{k-1} large enough to dominate the degrees of all irreducible components of an intersection of a degree- Δ_k variety with a degree- D_k hypersurface. The process ends after at most m steps. Let

$$R = \max_k \Delta_k.$$

All constants below are uniform for the finitely many dimensions and degree bounds in this tower.

Let W be a node of the resulting induction, let $k = \dim W$, and let $A \subset X \cap W$ have size M . We only need to treat nodes with $\alpha(W) \leq a$; nodes with larger intrinsic exponent will be bounded directly. Put

$$\Lambda = \Lambda_{a,R}(X; V).$$

Then $\lambda_W(A) \leq \Lambda$.

Apply polynomial partitioning to A at degree $D = D_k$. By [theorem 3.2](#),

$$(3.14) \quad E(Y) \leq C \left(D^{\sigma(W)} \Lambda M^2 + D^{2\sigma(W)} \sum_i E(A_i) \right).$$

The cells satisfy

$$(3.15) \quad |A_i| \leq C \frac{M}{D^k}, \quad \sum_i |A_i| \leq M.$$

By strong induction on cardinality,

$$E(A_i) \leq C_k \Lambda^{3-a} |A_i|^{a+\varepsilon}.$$

Consequently,

$$(3.16) \quad \begin{aligned} D^{2\sigma(W)} \sum_i E(A_i) &\leq C C_k \Lambda^{3-a} D^{2\sigma(W)} \left(\frac{M}{D^k} \right)^{a+\varepsilon-1} M \\ &= C C_k \Lambda^{3-a} M^{a+\varepsilon} D^{2\sigma(W)-k(a+\varepsilon-1)}. \end{aligned}$$

Since $a \geq \alpha(W)$,

$$a \geq 1 + \frac{2\sigma(W)}{k},$$

so the power of D in [\(3.16\)](#) is at most $-k\varepsilon$. The choice of D_k makes this contribution a fixed small fraction of the inductive target.

By [\(3.6\)](#),

$$(3.17) \quad D^{\sigma(W)} \Lambda M^2 \leq D^{\sigma(W)} \Lambda^{3-a} M^a.$$

For M above a fixed threshold, the missing factor M^ε absorbs the constant in (3.17); the remaining bounded range is included in the induction base.

It remains to control the wall

$$A_0 = A \cap W \cap Z(P).$$

By Bezout and irreducible decomposition, A_0 is contained in a union of $O_{n,\Delta_k,D_k}(1)$ admissible irreducible varieties U_j of dimension strictly less than k and degree at most Δ_{k-1} . Assign each wall point to one component and write $A_0 = \bigsqcup_j B_j$.

If $\alpha(U_j) \leq a$, the induction on dimension gives

$$E(B_j) \leq C_{k-1} \Lambda^{3-a} |B_j|^{a+\varepsilon}.$$

If $\alpha(U_j) > a$, the definition of Λ gives $|B_j| \leq \Lambda$, and the trivial estimate yields

$$(3.18) \quad E(B_j) \leq |B_j|^3 \leq \Lambda^{3-a} |B_j|^a \leq \Lambda^{3-a} |B_j|^{a+\varepsilon}.$$

Zero-dimensional components contain only $O_{n,R}(1)$ real points and are harmless. Using (2.5) and the bounded number of components,

$$(3.19) \quad E(A_0) \leq C'_{k-1} \Lambda^{3-a} M^{a+\varepsilon}.$$

Finally, $A = Y \sqcup A_0$, and (2.5) implies

$$E(A) \leq 8(E(Y) + E(A_0)).$$

Choose the dimension- k induction constant larger than the fixed wall constant and the bounded-cardinality base constant. The cellular term is contractive by (3.16). This completes both inductions and proves (1.8). $\square$

3.5. Consequences and recovery of the surface theorem.

Corollary 3.3. *Let V be admissible irreducible with $\alpha(V) < 3$, and put $a = \alpha(V)$. Then*

$$(3.20) \quad E(X) \ll_{V,\varepsilon} \Lambda_{a,R}(X; V)^{3-a} |X|^{a+\varepsilon}.$$

If the flag parameter is bounded independently of X , then

$$E(X) \ll_{V,\varepsilon} |X|^{\alpha(V)+\varepsilon}.$$

Corollary 3.4. *Let $V \subset \mathbb{R}^3$ be an admissible irreducible nonplanar algebraic surface of bounded degree. Then the choice $a = 2$ in theorem 1.1 gives*

$$(3.21) \quad E(X) \ll_{\deg V, \varepsilon} \Lambda_V(X) |X|^{2+\varepsilon}, \quad X \subset V,$$

where $\Lambda_V(X) = \max\{1, \sup_{\ell \subset V} |X \cap \ell|\}$ and the supremum is over affine lines contained in V .

Proof. We first note that $\dim \mathcal{D}(V) = 3$. Since a translate of V is contained in $V - V$, this dimension is at least two. If it were equal to two, then for generic smooth $x, y \in V$ the image of the differential of the difference map would be $T_x V + T_y V$ and would have dimension at most two. Hence $T_x V = T_y V$ for generic pairs. The tangent plane is therefore constant on a

dense open subset; every linear functional annihilating that plane is constant on V , so irreducibility places V in an affine plane, contrary to the hypothesis. Hence $\sigma(V) = 1$ and $\alpha(V) = 2$.

If $t \notin H_V$, the two irreducible hypersurfaces $V_{\mathbb{C}}$ and $V_{\mathbb{C}} + t$ have no common two-dimensional component, so there is no non-stabilizer excess fiber. The stabilizer has dimension at most one, since a two-dimensional stabilizer would make V an affine plane. Its nontrivial cosets are affine lines contained in V .

For a positive-dimensional proper subvariety of V , the only case with intrinsic exponent larger than two is an affine line. Indeed, if an irreducible curve had one-dimensional difference variety, the same tangent-space argument would force its tangent line to be constant, hence the curve would be affine. A nonlinear irreducible curve therefore has two-dimensional difference variety, zero-dimensional generic difference fibers, and no nonzero translation stabilizer. Bezout bounds all its nonzero translate intersections by a degree-dependent constant. Thus the flag parameter is comparable, with degree-dependent constants, to $\Lambda_V(X)$. Apply [theorem 1.1](#). $\square$

Remark 3.5. The proof above is not intended to replace the sharper and more direct wall analysis in [\[11\]](#). Its purpose is to show that their line parameter is the first member of a general translation–partition flag hierarchy.

4. A CODIMENSION-TWO QUADRATIC THREEFOLD

4.1. Normal form and difference fibers. Let

$$\Sigma(Q_1, Q_2) = \{(u, Q_1(u), Q_2(u)) : u \in \mathbb{R}^3\} \subset \mathbb{R}^5.$$

Invertible linear transformations preserve additive relations and energy. If Q_1 is positive definite and the generalized spectrum is simple, an invertible linear change in u and an invertible linear change in the last two target coordinates reduce the problem to

$$(4.1) \quad \Phi(u) = (u, |u|^2, u^T \Lambda u), \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \lambda_3), \quad \lambda_i \neq \lambda_j.$$

We work in this normal form.

Lemma 4.1. *The variety $\Sigma = \Phi(\mathbb{R}^3)$ contains no nontrivial affine line. More precisely, every affine line in $\mathbb{R}^5$ meets Σ in at most two points unless it is contained in Σ , and the latter alternative does not occur.*

Proof. If a line is contained in Σ , its projection onto the first three coordinates is either a point or a line $u_0 + sv$. The first case gives a constant line. In the second case, the coordinate $|u_0 + sv|^2$ has quadratic coefficient $|v|^2$, which cannot vanish for $v \neq 0$. Thus no nontrivial line is contained in Σ . The intersection bound follows from Bezout, or directly from the quadratic coordinate. $\square$

Let $A \subset \mathbb{R}^3$ be finite and put $X = \Phi(A)$. Fix a difference

$$t = \Phi(u) - \Phi(v) = (h, \tau_1, \tau_2), \quad h = u - v.$$

Then u satisfies

$$(4.2) \quad 2u \cdot h - |h|^2 = \tau_1,$$

$$(4.3) \quad 2u \cdot \Lambda h - h \cdot \Lambda h = \tau_2.$$

If h and Λh are linearly independent, these equations define an affine line in parameter space. Since the eigenvalues are distinct, dependence occurs exactly when

$$(4.4) \quad h \in \mathbb{R}e_1 \cup \mathbb{R}e_2 \cup \mathbb{R}e_3.$$

4.2. The rank-drop energy. Let $\mathcal{R}$ denote the set of differences whose parameter displacement belongs to the union in (4.4).

Lemma 4.2. *For every finite $A \subset \mathbb{R}^3$ and $X = \Phi(A)$,*

$$(4.5) \quad \sum_{t \in \mathcal{R}} r_X(t)^2 \leq 7|X|^2.$$

Proof. We treat the direction e_1 . Let R be the finite set of first coordinates occurring in A , let Z be the finite set of pairs (u_2, u_3) occurring in A , and define the incidence matrix

$$M_{r,z} = \mathbf{1}_A(r, z), \quad r \in R, z \in Z.$$

For $r \neq s$, the full difference associated with a pair $(r, z), (s, z)$ determines the ordered pair (r, s) , since

$$h = r - s, \quad r^2 - s^2 = h(r + s).$$

Its multiplicity is therefore

$$G_{r,s} = \sum_z M_{r,z} M_{s,z}, \quad G = M M^\top.$$

Hence the nonzero e_1 -direction contribution is at most

$$\sum_{r \neq s} G_{r,s}^2 \leq \text{tr}(G^2) \leq \text{tr}(G)^2 = |A|^2,$$

because G is positive semidefinite and $\text{tr}(G) = |A|$. The zero difference contributes $|A|^2$. Thus the e_1 family contributes at most $2|A|^2$. The same argument applies to e_2 and e_3 . Counting the zero difference only once gives (4.5). $\square$

4.3. A finite Freiman projection for wall surfaces. We need a version of the Jing–Wu theorem for a finite subset of a surface in an arbitrary ambient space.

Lemma 4.3. *Let $S \subset \mathbb{R}^M$ be an admissible irreducible algebraic surface of degree at most B that contains no affine line, and let $A \subset S$ be finite. There is a linear map*

$$L : \mathbb{R}^M \longrightarrow \mathbb{R}^3$$

such that:

- (i) L is a Freiman isomorphism of order two on A ;

- (ii) the Zariski closure $S' = \overline{L(S)}^{\text{Zar}}$ is an irreducible nonplanar algebraic surface of degree $O_B(1)$;
- (iii) every affine line in S' contains at most B points of $L(A)$.

Consequently,

$$(4.6) \quad E(A) \ll_{B,\varepsilon} |A|^{2+\varepsilon}.$$

Proof. Choose L generically. The failure of the Freiman property means that $\ker L$ contains a nonzero vector from the finite set

$$(A + A - A - A) \setminus \{0\},$$

so this is avoided by a generic choice. We also require that every noncollinear triple in A remain noncollinear after projection; each failure is a proper algebraic condition on L .

The linear maps for which the image dimension drops, the affine span of the image is planar, or the generic projection degree/degree bound fails form a proper algebraic exceptional set. Thus a generic linear projection of an irreducible surface has two-dimensional irreducible image closure and degree bounded in terms of B ; this is the standard generic-projection theorem, together with the degree bound for a linear image. Since S contains no affine plane, its affine span has dimension at least three, and L may be chosen so that the image is not planar.

Suppose an affine line $\ell \subset S'$ contains more than B points of $L(A)$. By preservation of noncollinearity, their preimages lie on one affine line in $\mathbb{R}^M$. That line meets S in more than B points, so Bezout forces it to be contained in S , a contradiction. Thus the line concentration of $L(A)$ is at most B . Since L preserves all additive quadruples, $E(A) = E(L(A))$, and (4.6) follows from (1.3). $\square$

For curves, an even simpler observation suffices.

Lemma 4.4. *Let $C \subset \mathbb{R}^M$ be an admissible irreducible algebraic curve of degree at most B that is not an affine line. Then every finite $A \subset C$ satisfies*

$$(4.7) \quad E(A) \ll_B |A|^2.$$

Proof. If $t \neq 0$ and $C = C + t$, then the translation stabilizer contains a line, forcing C itself to be an affine line. Hence C and $C + t$ are distinct for every $t \neq 0$, and Bezout gives

$$r_A(t) \leq |C \cap (C + t)| \ll_B 1.$$

The zero difference contributes $|A|^2$, and

$$\sum_{t \neq 0} r_A(t)^2 \ll_B \sum_t r_A(t) = O_B(|A|^2).$$

$\square$

4.4. Polynomial partitioning in parameter space. Apply the three-dimensional polynomial partitioning theorem to $A \subset \mathbb{R}^3$ at degree D . Let

$$A_0 = A \cap Z(q), \quad A \setminus A_0 = \bigsqcup_i A_i, \quad X_i = \Phi(A_i), \quad Y = \bigsqcup_i X_i.$$

Then

$$(4.8) \quad |A_i| \ll \frac{|A|}{D^3}, \quad \#\{i : A_i \neq \emptyset\} \ll D^3.$$

Proposition 4.5. *With the notation above,*

$$(4.9) \quad E(Y) \ll D|X|^2 + D^2 \sum_i E(X_i).$$

Proof. For a difference $t \notin \mathcal{R}$, the solution set of (4.2)–(4.3) is an affine line L_t in parameter space. The restrictions of $q(u)$ and $q(u-h)$ to this line have degree at most D . If either restriction vanishes identically, there is no contribution from Y on the corresponding side. Otherwise their zero sets cut L_t into $O(D)$ intervals. On each interval, u and $u-h$ remain in fixed partition cells. Hence

$$(4.10) \quad \#\{(i, j) : r_{X_i - X_j}(t) > 0\} \ll D, \quad t \notin \mathcal{R}.$$

Cauchy–Schwarz and [theorem 4.2](#) give

$$(4.11) \quad E(Y) \ll |X|^2 + D \sum_{i,j} \sum_t r_{X_i - X_j}(t)^2.$$

By (2.4),

$$\sum_{i,j} \sum_t r_{X_i - X_j}(t)^2 = \sum_s \left(\sum_i r_{X_i - X_i}(s) \right)^2.$$

For $s \notin \mathcal{R}$, (4.10) with $i = j$ gives at most $O(D)$ nonzero summands. The contribution from $s \in \mathcal{R}$ is bounded by [theorem 4.2](#). Therefore

$$(4.12) \quad \sum_{i,j} \sum_t r_{X_i - X_j}(t)^2 \ll |X|^2 + D \sum_i E(X_i).$$

Substitution into (4.11) proves the proposition. $\square$

4.5. The wall and the proof of the theorem. Let

$$W = \Phi(A_0).$$

Factor q and decompose A_0 among the real irreducible components of its zero set. Under the polynomial embedding Φ , each positive-dimensional component becomes an algebraic subvariety of Σ of dimension at most two and degree $O_D(1)$. By [theorem 4.1](#), none contains an affine line. Two-dimensional components satisfy (4.6); one-dimensional components satisfy (4.7); zero-dimensional components contain $O_D(1)$ points. The fourth-root union inequality therefore gives, for every $\eta > 0$,

$$(4.13) \quad E(W) \ll_{D,\eta} |W|^{2+\eta}.$$

Proof of theorem 1.2. We argue by strong induction on $N = |X|$. Take $\eta = \varepsilon/2$ and fix a sufficiently large constant degree $D = D(\varepsilon)$. By (2.5), theorem 4.5, and (4.13),

$$(4.14) \quad E(X) \ll DN^2 + D^2 \sum_i E(X_i) + C_{D,\varepsilon} N^{2+\varepsilon/2}.$$

Using the inductive hypothesis and (4.8),

$$(4.15) \quad \begin{aligned} D^2 \sum_i E(X_i) &\ll C_\varepsilon D^2 \left(\frac{N}{D^3} \right)^{1+\varepsilon} \sum_i |X_i| \\ &\ll C_\varepsilon D^{-1-3\varepsilon} N^{2+\varepsilon}. \end{aligned}$$

Choose D so that the coefficient of $C_\varepsilon N^{2+\varepsilon}$ is less than $1/4$. The first and third terms in (4.14) are absorbed for N above a fixed threshold, and the remaining finite range is included in the induction constant. This proves (1.9).

The diagonal solutions give $E(X) \geq 2N^2 - N$, so the exponent is sharp. $\square$

Remark 4.6. If $Q_2 = \lambda Q_1$, then $\Sigma(Q_1, Q_2)$ lies in an affine hyperplane and is linearly equivalent to a three-dimensional paraboloid in $\mathbb{R}^4$. Lattice boxes on that paraboloid have energy of order $N^{7/3}$; see the discussion surrounding the paraboloid threshold in [12]. Thus a rank condition on the quadratic pencil is genuinely necessary for a universal $N^{2+\varepsilon}$ theorem. The intermediate multiplicity pattern $2 + 1$ should admit a refinement using lower-dimensional paraboloid fibers, but it is not treated here.

5. WEIGHTED AND OFF-DIAGONAL YOUNG INEQUALITIES

5.1. From hereditary energy to weighted L^4 . We first record an abstract extension principle.

Proposition 5.1. *Let X be finite, let $2 \leq a < 3$, and let $K \geq 1$. Suppose that for every $\eta > 0$ there is C_η such that*

$$(5.1) \quad E(A) \leq C_\eta K^{3-a} |A|^{a+\eta} \quad (A \subset X).$$

Then, for every $\varepsilon > 0$, there is $\eta = \eta(a, \varepsilon) > 0$ such that every f supported on X satisfies

$$(5.2) \quad \|\widehat{f}\|_{L^4} \ll_{a,\varepsilon} C_\eta^{1/4} K^{(3-a)/4} |X|^\varepsilon \|f\|_{\ell^{4/a}}.$$

Proof. Choose $\eta = \eta(a, \varepsilon) > 0$ sufficiently small and put

$$\frac{1}{p_\eta} = \frac{a + \eta}{4}.$$

The indicator estimate (5.1) says that the Fourier extension operator is of restricted type $(p_\eta, 4)$ with norm $O(C_\eta^{1/4} K^{(3-a)/4})$. The standard restricted-type extension (equivalently, a dyadic decomposition of the coefficients) gives

$$(5.3) \quad \|\widehat{f}\|_{L^4} \ll C_\eta^{1/4} K^{(3-a)/4} \|f\|_{\ell^{p_\eta,1}}.$$

For a sequence supported on at most $N = |X|$ points,

$$\|f\|_{\ell^{p_\eta,1}} \ll_{p_\eta} (\log(2N))^{1-1/p_\eta} \|f\|_{\ell^{p_\eta}}.$$

Since $p_\eta < 4/a$,

$$\|f\|_{\ell^{p_\eta}} \leq N^{\eta/4} \|f\|_{\ell^{4/a}}.$$

Choose η sufficiently small in terms of ε and absorb the logarithm into N^ε . This proves the claim. $\square$

Applying [theorem 1.1](#) to every subset of X is legitimate because the flag parameter is monotone. We therefore obtain the following.

Corollary 5.2. *In the setting of [theorem 1.1](#), let $\Lambda = \Lambda_{a,R}(X; V)$. Then*

$$(5.4) \quad \|\widehat{f}\|_{L^4} \ll_{V,a,\varepsilon} \Lambda^{(3-a)/4} |X|^\varepsilon \|f\|_{\ell^{4/a}}, \quad \text{supp } f \subset X.$$

For the quadratic threefold, [theorem 1.2](#) gives

$$(5.5) \quad \|\widehat{f}\|_{L^4} \ll_\varepsilon |X|^\varepsilon \|f\|_{\ell^2}.$$

5.2. The off-diagonal interpolation polygon. Let X, Y satisfy hereditary estimates with exponents a_X, a_Y and parameters K_X, K_Y . Put

$$p_X = \frac{4}{a_X}, \quad p_Y = \frac{4}{a_Y}.$$

The geometric endpoint obtained from (2.2) and [theorem 5.1](#) is

$$(5.6) \quad \|f * g\|_{\ell^2} \ll (K_X^{3-a_X} K_Y^{3-a_Y})^{1/4} (|X||Y|)^\varepsilon \|f\|_{\ell^{p_X}} \|g\|_{\ell^{p_Y}}.$$

The classical endpoints are

$$(5.7) \quad \|f * g\|_{\ell^2} \leq \|f\|_{\ell^1} \|g\|_{\ell^2}, \quad \|f * g\|_{\ell^2} \leq \|f\|_{\ell^2} \|g\|_{\ell^1}.$$

Theorem 5.3. *Let X, Y be finite sets. Suppose that, for every $\eta > 0$ and all subsets $A \subset X, B \subset Y$,*

$$E(A) \ll_\eta K_X^{3-a_X} |A|^{a_X+\eta}, \quad E(B) \ll_\eta K_Y^{3-a_Y} |B|^{a_Y+\eta},$$

where $2 \leq a_X, a_Y < 3$ and $K_X, K_Y \geq 1$. Let $\theta_0, \theta_1, \theta_2 \geq 0$ with $\theta_0 + \theta_1 + \theta_2 = 1$, and define

$$(5.8) \quad \frac{1}{p} = \theta_0 \frac{a_X}{4} + \theta_1 + \frac{\theta_2}{2},$$

$$(5.9) \quad \frac{1}{q} = \theta_0 \frac{a_Y}{4} + \frac{\theta_1}{2} + \theta_2.$$

Then

$$(5.10) \quad \|f * g\|_{\ell^2} \ll_\varepsilon K_X^{\theta_0(3-a_X)/4} K_Y^{\theta_0(3-a_Y)/4} (|X||Y|)^\varepsilon \|f\|_{\ell^p} \|g\|_{\ell^q}.$$

The same estimate holds after decreasing either p or q .

Proof. Apply bilinear complex interpolation to (5.6) and the two estimates in (5.7); see, for example, [5, 9]. The final assertion follows from monotonicity of counting-measure norms. $\square$

When $a_X = a_Y = 2$, the interpolation polygon and its upward monotone closure admit a simple description.

Corollary 5.4. *Let $V, W \subset \mathbb{R}^n$ be admissible irreducible bounded-degree varieties. Choose target exponents $a_V, a_W \in [2, 3)$ with $\alpha(V) \leq a_V$ and $\alpha(W) \leq a_W$, and let R_V, R_W be the degree cutoffs supplied by [theorem 1.1](#). For finite $X \subset V, Y \subset W$, put*

$$\Lambda_X = \Lambda_{a_V, R_V}(X; V), \quad \Lambda_Y = \Lambda_{a_W, R_W}(Y; W).$$

If $\theta_0, \theta_1, \theta_2$ and p, q satisfy (5.8)–(5.9) with $a_X = a_V$ and $a_Y = a_W$, then

$$\|f * g\|_{\ell^2} \ll_{n, \deg V, \deg W, p, q, \varepsilon} \Lambda_X^{\theta_0(3-a_V)/4} \Lambda_Y^{\theta_0(3-a_W)/4} (|X||Y|)^\varepsilon \|f\|_{\ell^p} \|g\|_{\ell^q}.$$

The estimate remains valid after decreasing either input exponent.

Proof. Apply [theorems 5.2](#) and [5.3](#) and use the monotonicity of the flag parameters on subsets. $\square$

Corollary 5.5. *Assume the hypotheses of [theorem 1.3](#). Then (1.11) holds.*

Proof. For

$$1 \leq \frac{1}{p} + \frac{1}{q} \leq \frac{3}{2},$$

set

$$\theta_0 = 3 - 2 \left(\frac{1}{p} + \frac{1}{q} \right).$$

The remaining barycentric mass can be split between the two classical endpoints because $1/2 \leq 1/p, 1/q \leq 1$. If $1/p + 1/q \geq 3/2$, use the classical Young line and norm monotonicity, so $\theta_0 = 0$. $\square$

5.3. Applications and sharpness.

Corollary 5.6. *Let $X \subset V$ and $Y \subset W$, where $V, W \subset \mathbb{R}^3$ are admissible irreducible nonplanar algebraic surfaces of bounded degree. For $1 \leq p, q \leq 2$ with $1/p + 1/q \geq 1$, let*

$$\theta = \max \left\{ 0, 3 - 2 \left(\frac{1}{p} + \frac{1}{q} \right) \right\}.$$

Then

$$(5.12) \quad \|f * g\|_{\ell^2} \ll_{\deg V, \deg W, p, q, \varepsilon} (\Lambda_V(X) \Lambda_W(Y))^{\theta/4} (|X||Y|)^\varepsilon \|f\|_{\ell^p} \|g\|_{\ell^q}.$$

Corollary 5.7. *Let X, Y lie on quadratic threefolds satisfying the hypotheses of [theorem 1.2](#). Then, for every $1 \leq p, q \leq 2$ with $1/p + 1/q \geq 1$,*

$$(5.13) \quad \|f * g\|_{\ell^2} \ll_{p, q, \varepsilon} (|X||Y|)^\varepsilon \|f\|_{\ell^p} \|g\|_{\ell^q}.$$

The exponent region in [theorem 5.7](#) is optimal up to the subpolynomial loss. Indeed, taking g to be a point mass forces $p \leq 2$, and symmetrically

$q \leq 2$. Taking $f = g = \mathbf{1}_X$ and using the diagonal lower bound $E(X) \gtrsim |X|^2$ forces

$$\frac{1}{p} + \frac{1}{q} \geq 1.$$

The concentration power in [theorem 5.6](#) is also sharp in the joint sense. Let $X = Y$ be an arithmetic progression of length N on a line contained in the surface. Then

$$\|\mathbf{1}_X * \mathbf{1}_X\|_{\ell^2} \asymp N^{3/2}, \quad \Lambda_V(X) = N.$$

When $1 \leq 1/p + 1/q \leq 3/2$, the right side of [\(5.12\)](#) has N -exponent

$$\frac{1}{p} + \frac{1}{q} + \frac{\theta}{2} = \frac{3}{2}.$$

Thus the displayed power of the product concentration parameter cannot be uniformly reduced.

5.4. Critical exponents from dimension. If $V^m \subset \mathbb{R}^{m+c}$ has maximal difference dimension, then [\(1.7\)](#) and [theorem 5.2](#) give the critical symmetric input exponent

$$(5.14) \quad p_c(m, c) = \frac{4}{\alpha_{m,c}} = \begin{cases} 2, & m \leq 2c, \\ \frac{4m}{3m-2c}, & m > 2c. \end{cases}$$

Thus codimension two gives $p_c = 2$ for $m = 3, 4$, while $p_c = 20/11$ for $m = 5$. At exponents $a > 2$, [theorem 5.3](#) supplies the interpolation polygon generated by the symmetric geometric endpoint and the two classical Young endpoints. Determining a larger genuinely asymmetric sharp region would require new mixed-energy information and is not a consequence of self-energy alone.

6. TURNING COMPLEXITY AND THREE-FOLD ENERGY ON CURVES

6.1. Signed-convex complexity. Let $P \subset \mathbb{R}^2$ be finite and let ℓ be a linear functional that is injective on P . Choose a linear functional m linearly independent of ℓ and enumerate

$$P = \{p_1, \dots, p_N\}, \quad \ell(p_1) < \dots < \ell(p_N).$$

Put

$$(6.1) \quad s_i = \frac{m(p_{i+1}) - m(p_i)}{\ell(p_{i+1}) - \ell(p_i)}, \quad 1 \leq i < N.$$

A consecutive point block $\{p_a, \dots, p_b\}$ is *signed-convex* if $s_a, \dots, s_{b-1}$ is strictly increasing or strictly decreasing. Let $\kappa_\ell(P)$ be the least number of consecutive signed-convex blocks in a partition of P .

If m' is another complementary coordinate, then $m' = cm + d\ell$ with $c \neq 0$, so the new slopes are $cs_i + d$. Thus $\kappa_\ell(P)$ is independent of m . One may define the affine-invariant complexity

$$\kappa(P) = \min_{\ell} \kappa_\ell(P),$$

where the minimum is over projections injective on P .

Lemma 6.1. *If the adjacent slopes of a finite graph data set*

$$x_1 < \cdots < x_M, \quad (x_i, y_i) \in \mathbb{R}^2$$

are strictly increasing, then there is a strictly convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x_i) = y_i$ for every i . If the slopes are strictly decreasing, there is a strictly concave interpolant.

Proof. We treat the increasing case. Let

$$\sigma_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}.$$

Choose derivatives d_i so that

$$d_1 < \sigma_1, \quad \sigma_{i-1} < d_i < \sigma_i \quad (2 \leq i \leq M-1), \quad d_M > \sigma_{M-1}.$$

On each interval $[x_i, x_{i+1}]$, choose a continuous strictly increasing function g_i with endpoint values d_i, d_{i+1} and average σ_i . Such a function is obtained by moving the transition point in a monotone piecewise-linear profile; it may be smoothed without changing the endpoint values or the integral. Define

$$f(x) = y_i + \int_{x_i}^x g_i(t) dt \quad (x \in [x_i, x_{i+1}]).$$

Then f is C^1 , has strictly increasing derivative, and interpolates the data. Extend the derivative monotonically outside $[x_1, x_M]$. The decreasing case follows by replacing y with $-y$. $\square$

6.2. A strip projection estimate.

Lemma 6.2. *Let $g_1, \dots, g_K \in L^6(\mathbb{R}^2)$ have Fourier supports contained in pairwise disjoint strips of the form*

$$\{\xi : a_j < \ell(\xi) < b_j\},$$

with disjoint intervals (a_j, b_j) . For every $\delta > 0$,

$$(6.2) \quad \left\| \sum_{j=1}^K g_j \right\|_{L^6}^6 \ll_{\delta} K^{4+\delta} \sum_{j=1}^K \|g_j\|_{L^6}^6.$$

Proof. After a linear change of coordinates, take $\ell(\xi) = \xi_1$. Let P_j be the Fourier projection to the j th strip and define

$$T((h_j)_j) = \sum_j P_j h_j.$$

Orthogonality gives

$$T : \ell^2(L^2) \longrightarrow L^2$$

with norm one. For every $r > 6$, interval projections in one Fourier coordinate are uniformly bounded on L^r by the Hilbert-transform theorem, and therefore

$$\|T(h_j)\|_{L^r} \ll_r \sum_j \|h_j\|_{L^r} \leq K^{1-1/r} \left(\sum_j \|h_j\|_{L^r}^r \right)^{1/r}.$$

Interpolate these vector-valued estimates at the parameter θ determined by

$$\frac{1}{6} = \frac{1-\theta}{2} + \frac{\theta}{r}.$$

This gives

$$\|T(h_j)\|_{L^6} \ll_r K^{\theta(1-1/r)} \left(\sum_j \|h_j\|_{L^6}^6 \right)^{1/6}.$$

A calculation yields

$$6\theta \left(1 - \frac{1}{r} \right) = \frac{4(r-1)}{r-2} = 4 + \frac{4}{r-2}.$$

Choose r so that $4/(r-2) < \delta$. □

6.3. Admissible thickening. For a finite $P \subset \mathbb{R}^2$, let

$$\mathcal{S}_3(P) = \{p_1 + p_2 + p_3 - p_4 - p_5 - p_6 : p_i \in P\}.$$

Choose a Schwartz function ϕ whose Fourier transform is supported in a sufficiently small ball and normalize $\int |\phi|^6 = 1$. If the support is small enough that

$$\text{supp } \widehat{|\phi|^6} \cap (\mathcal{S}_3(P) \setminus \{0\}) = \emptyset,$$

then

$$(6.3) \quad \|\phi(x) \sum_{p \in P} e^{2\pi i p \cdot x}\|_{L^6}^6 = J_3(P).$$

This is the admissible thickening used in [8]. Since P is finite, the Fourier support of ϕ may simultaneously be chosen small enough that different consecutive point blocks occupy disjoint ℓ -strips.

6.4. Proof of the complexity theorem.

Proof of [theorem 1.4](#). Fix a partition into $K = \kappa_\ell(P)$ consecutive signed-convex blocks

$$P = P_1 \sqcup \dots \sqcup P_K.$$

Write $n_j = |P_j|$. The invertible coordinate map $p \mapsto (\ell(p), m(p))$ preserves all additive relations. By [theorem 6.1](#) and the theorem of Cushman–Demeter–Wu [8], for every $\eta > 0$,

$$(6.4) \quad J_3(P_j) \ll_\eta n_j^{3+\eta}.$$

For a common admissible thickening, define

$$F_j(x) = \phi(x) \sum_{p \in P_j} e^{2\pi i p \cdot x}.$$

Their Fourier supports lie in pairwise disjoint strips, and

$$J_3(P) = \left\| \sum_j F_j \right\|_{L^6}^6, \quad \|F_j\|_{L^6}^6 = J_3(P_j).$$

The block sizes may be uneven. Group them dyadically: for $r \geq 0$, let

$$\mathcal{J}_r = \{j : 2^r \leq n_j < 2^{r+1}\}, \quad K_r = |\mathcal{J}_r|, \quad N_r = \sum_{j \in \mathcal{J}_r} n_j,$$

and put $F^{(r)} = \sum_{j \in \mathcal{J}_r} F_j$. By [theorem 6.2](#) and (6.4),

$$\begin{aligned} \|F^{(r)}\|_{L^6}^6 &\ll_{\eta} K_r^{4+\eta} \sum_{j \in \mathcal{J}_r} n_j^{3+\eta} \\ (6.5) \quad &\ll_{\eta} K_r^{5+\eta} 2^{r(3+\eta)} \asymp K_r^2 N_r^{3+\eta}. \end{aligned}$$

Hence

$$\|F^{(r)}\|_{L^6} \ll_{\eta} K^{1/3} N_r^{1/2+\eta/6}.$$

There are at most $1 + \log_2 N$ nonempty classes. Minkowski and Hölder in the class index give

$$\begin{aligned} J_3(P)^{1/6} &\leq \sum_r \|F^{(r)}\|_{L^6} \\ (6.6) \quad &\ll_{\eta} K^{1/3} (1 + \log N)^{1/2-\eta/6} N^{1/2+\eta/6}. \end{aligned}$$

Taking sixth powers and choosing $\eta < \varepsilon$ proves

$$J_3(P) \ll_{\varepsilon} K^2 N^{3+\varepsilon}.$$

□

6.5. Sharp examples.

Proposition 6.3. *For every pair of integers $K, n \geq 1$, one can find a set $P \subset \mathbb{R}^2$ with $|P| = Kn$ and an injective projection ℓ such that*

$$\kappa_{\ell}(P) \asymp K, \quad J_3(P) \asymp K^2 |P|^3.$$

Proof. Set $N = Kn$. Fix an integer $B \geq 4$ and put

$$a_i = B^i, \quad 1 \leq i \leq n.$$

Choose $H > 3n$ and $C = a_n - a_1$, and define

$$(6.7) \quad P_{K,n} = \{(Hj + i, Cj + a_i) : 0 \leq j < K, 1 \leq i \leq n\}.$$

Within each packet, the adjacent slopes are

$$a_{i+1} - a_i = (B - 1)B^i$$

and are strictly increasing. The slope joining two consecutive packets is zero. Thus the slope sequence has $K - 1$ strict resets and

$$\kappa_\ell(P_{K,n}) \asymp K$$

for the first-coordinate projection.

Suppose two ordered triples in $P_{K,n}$ have the same sum. The first coordinate gives

$$H\Delta J + \Delta I = 0, \quad |\Delta I| < 3n < H,$$

so $\Delta J = \Delta I = 0$. The second coordinate then gives $\Delta A = 0$. Consequently the equality separates into an equality of three-term sums in $\{0, \dots, K - 1\}$ and an equality of three-term vector sums in

$$B_n = \{(i, B^i) : 1 \leq i \leq n\}.$$

Because $B \geq 4$, a sum of three powers B^i uniquely determines the multiset of exponents. Hence

$$J_3(B_n) \asymp n^3.$$

Also

$$J_3(\{0, \dots, K - 1\}) \asymp K^5.$$

Therefore

$$J_3(P_{K,n}) \asymp K^5 n^3 = K^2 N^3.$$

□

Choosing integer sequences with $K \asymp N^\beta$ gives the full sharp power spectrum

$$J_3(P) \asymp N^{3+2\beta}, \quad 0 \leq \beta \leq 1.$$

7. HIGHER-DIMENSIONAL CONSEQUENCES, SHARPNESS, AND LIMITATIONS

7.1. Maximal difference dimension. If $V^m \subset \mathbb{R}^{m+c}$ has maximal difference dimension, then (1.7) gives

$$(7.1) \quad \alpha_{m,c} = \max \left\{ 2, 3 - \frac{2c}{m} \right\}.$$

Thus

(m, c)	σ	$\alpha_{m,c}$
$(2, 1)$	1	2
$(3, 2)$	1	2
$(4, 2)$	2	2
$(3, 1)$	2	7/3
$(5, 2)$	3	11/5

The cases $m \leq 2c$ are near-diagonal; $m = 2c$ is critical for the N^ε iteration; and $m < 2c$ is subcritical, as illustrated by the genuine D^{-1} gain in [theorem 1.2](#).

7.2. Quadratic lower bounds. Let $Q_1, \dots, Q_c$ be fixed integer quadratic forms on $\mathbb{R}^m$ and consider

$$V_Q = \{(u, Q_1(u), \dots, Q_c(u)) : u \in \mathbb{R}^m\}.$$

For

$$X_M = \{(u, Q_1(u), \dots, Q_c(u)) : u \in \{1, \dots, M\}^m\}, \quad N = M^m,$$

one has

$$|X_M - X_M| \ll M^{m+2c}.$$

Cauchy–Schwarz therefore gives

$$(7.2) \quad E(X_M) \geq \frac{N^4}{|X_M - X_M|} \gg N^{3-2c/m}.$$

For a generic choice of the forms Q_j , the difference map has maximal image dimension. Together with the diagonal contribution, this shows that no uniform theorem for all bounded-degree varieties in the maximal-difference class can improve the exponent in (7.1). Special varieties can improve on the universal exponent: in four dimensions, sphere geometry breaks the $7/3$ threshold, while the paraboloid realizes it; see Mudgal [12].

7.3. Sharpness of the flag power. Suppose a variety V_0 admits finite sets B with bounded base flag parameter and

$$E(B) \asymp |B|^a.$$

Let H be a one-dimensional affine factor and let $A \subset H$ be an arithmetic progression of length L . On the cylinder $V = V_0 \times H$, take $X = B \times A$. Then

$$E(X) = E(B)E(A) \asymp |B|^a L^3.$$

Writing $N = |X| = L|B|$, the flag parameter is comparable to L under the stated bounded-base hypothesis, and one obtains

$$(7.3) \quad E(X) \asymp L^{3-a} N^a.$$

Thus the power $3 - a$ in theorem 1.1 is forced whenever the base exponent is attained with bounded base flag complexity. In particular, a generic finite subset of a parabola, tensored with an arithmetic progression, gives the sharp case $a = 2$.

7.4. What the flag theorem does not remove. The flag theorem handles arbitrary bounded-degree varieties by charging the maximum occupancy of exceptional translate intersections. This is sometimes too crude. In the quadratic threefold, the rank-drop planes may contain many selected points, but theorem 4.2 shows that their total difference energy is only quadratic. A sharper general theorem would replace the maximum flag occupancy by an aggregate quantity such as

$$(7.4) \quad \mathfrak{G}_W(X) = \frac{1}{|X|^2} \sum_{t: \dim(W \cap (W+t)) > \sigma(W)} r_X(t)^2.$$

Controlling (7.4) requires additional geometry of the critical locus of the difference map. For quadratic graphs it becomes a determinantal rank problem; for general varieties it is related to secant and contact defects and to translation rulings. This is the principal obstruction to a parameter-free theorem for every bounded-degree variety.

A second open direction concerns the off-diagonal region when $\alpha(V) > 2$. Self-energy supplies a symmetric $\ell^{4/\alpha} \rightarrow L^4$ endpoint, and theorem 5.3 gives its interpolation polygon. A larger sharp asymmetric region would require genuinely mixed geometric energy estimates rather than Cauchy–Schwarz between two self-energies.

8. STATEMENT ON THE USE OF AI

AI systems were used extensively during exploratory work and preparation. ChatGPT 5.6 Pro assisted with the discovery and checking of the difference-fiber recurrence, the flag formulation, the codimension-two quadratic model, the weighted interpolation argument, the turning-complexity examples, literature searches, exposition, and LaTeX preparation. The author retains responsibility for independently verifying every argument and for all final mathematical and expository decisions.

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